Requirements Traceability Matrix

ID	Requirement	Related Use Case	Test	Implementation Element / Fulfilled By	Description
1	Power ON/OFF device with button press, and secure PIN entry	Insulin Pump Step 1.1 (a-c) and Ext-1c	Run simulator in Qt and click power ON/OFF and enter PIN. Try out different combinations.	batterymanager.cpp , config.cpp	User sets PIN on device's first power up, is asked to verify pin on each power up thereafter, power off turns off device and hides UI, locks access until correct pin is entered.
2	Battery/Charger widget and low-battery alert	Ext-1b	Let the battery drain, wait to ensure that the 'low battery warning' is displayed, and then click the 'plug in' button to charge the device.	BatteryManager singleton	Visual battery gauge drains 1% every 3 s while unplugged and charges 1% every second when plugged. At 10%, displays a warning. Forces shutdown at 0%.
3	Home screen shows real-time CGM graph and stats	Insulin Pump Step 1.2	Run simulator in qt to see the graph's readings, wait 10 seconds to see new reading.	glucosemonitoring. cpp, mainwindow.cpp (homescreen widget)	CGM linear graph on pump's home page updates every 10 seconds of simulated time and always shows latest BG value.
4	Create / Edit /	Insulin Pump Step	Select profiles and	personalprofiles.cp	Users are able to create

	Delete personal profiles for user	1.2 (a-c)	try out the profile creation, deletion, edit	p, profile.h, optionsmenu.cpp, timedsettingprofiles .cpp, timedSettingsUtils.c pp, profiles.json, timedSettings.json	/ update / delete profiles with basal, carb ratio, etc. They can edit, rename, and the changes persist to disk (profiles.json)
5	Manual bolus from user or CGM	Insulin Pump Step 1.3	Click on bolus and enter values such as the Carbohydrates if that's turned on in the active personal profile, enter the current blood glucose amount unless auto populated and click calculate bolus button.	Boluscalculator.cpp ,bolusmenu.cpp , glucosemonitoring. cpp	From the bolus page, users can enter carbs + current BG. Bolus calculator applies formula from User Guide to recommend a dose; user can also override dosage. Result is logged.
6	Extended / quick bolus split and cancel mid-delivery	Ext-3a/3b	Press the extended bolus button and fill in the box of “deliver now” and “deliver later”	Boluscalculator.cpp ,bolusmenu.cpp	Calculator can split dose into immediate portion / extended portion delivered overtime. Log records units delivered.
7	Start / stop / resume basal (Control-IQ)	Insulin Pump Step 1.4 & Ext-2b/2c	Wait for reading generated than < 3.9 mmol/L to show pop up “low glucose alert”	Glucosemonitoring. cpp, optionsmenu.cpp	Basal is stopped when CGM detects hypoglycemia (< 3.9 mmol/L) and resumes when safe. Users can enable or disable Control-IQ in the options menu.

8	Log pump history (basal, bolus, alerts) 90 days	Insulin Pump Step 1.5-a, b	Press options menu and select history to check bolus, cgm and alerts.	Optionsmenu.cpp, historylogger.cpp	Users can view totals: units delivered, basal vs bolus, average blood glucose as well as low or high glucose warnings. All logs include a timestamp, as well as a notes section, and are displayed in the history screen, accessible from the options menu.
9	Navigation among pages with Tandem logo and stacked widget	N/A	Run the simulator and navigate	mainwindow.cpp	User can switch views, each pages maintains dimension, and global battery/clock status. Tandem “T” logo brings user back to home screen.